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No. of Printed Pages : 02

SARDAR PATEL UNIVERSITY

M.Sc.(Information Technology) FIRST Semester Examination

Subject : PS01CINT23: RDBMS & Client Server Computing

21/11/2019, Thursday

Time: 10 AM to 1 PM

Marks: 70

Q.1 Multiple Choice Questions.

[08]

- 1 _____ command is used to modify the structure of a table.
A. Modify B. Update C. Alter table D. None of given.
- 2 A _____ is a database object that holds user data.
A. Query B. Form C. Table D. None of given
- 3 _____ Section is used in PL/SQL to handle errors.
A. Error B. Handle error C. Exception D. None of given
- 4 _____ Command is used to give permission to other user to use table.
A. Permit B. Grant C. Revoke D. None of given
- 5 _____ Normalization process focus on transitive dependency.
A. 1NF B. 2NF C. 3NF D. None of given
- 6 _____ is the process of extracting the knowledge or design blueprints from anything man-made.
A. Forward Engineering B. Reengineering
C. Reverse Engineering D. None of given
- 7 In a _____ distributed database system, all databases are of different type.
A. Client/Server B. Homogeneous C. Heterogeneous D. None of given
8. In phase 3 of 3PC, coordinator sends _____ message to all participating sites.
A. commit/abort B. partitioning C. decision D. None of given

Q.2 Answer the following Short Questions. (Attempt any seven)

[14]

- 1 Differentiate clearly Trigger and stored procedure.
- 2 Write the uses of Database Trigger.
- 3 Write the difference between logical data independence and physical data independence.
- 4 Explain with example the built-in function ROUND().
- 5 Explain briefly with example the table DUAL.
- 6 Define terms: Heterogeneous services, Transparent gateway agents.
- 7 Distinguish between Procedural Programming & Event-driven Programming.
- 8 What is a relationship? Explain types of relationship.
- 9 Define E-R Diagrams and write its uses.

①

(P.T.O)

Q.3 Do as Directed.

[A] What is RDBMS? Explain in brief the Relational Data Model with advantages and disadvantages. [06]

[B] Define the term Normalization. Explain up to 3NF with example. [06]

OR

[B] List out and explain briefly the relational algebraic operations Relational database. [06]

Q.4 Do as Directed.

[A] Explain 2-tier & 3-tier client server architecture with diagrams. [06]

[B] What is event-driven programming? Explain how event-driven programming works. [06]

OR

[B] Explain Homogeneous Distributed Database System with example. [06]

Q.5 Do as Directed.

[A] Explain with example PL/SQL code block. [06]

[B] List out all rules of Dr. E.F.Codd's. Explain briefly any two of them. [06]

OR

[B] What do you mean by concurrency control? How to handle concurrency control? [06]

Q.6 Do as Directed.

[A] Write SQL command to perform following task. [Any THREE] [06]

- 1 To create a table EMP having fields (ENO number(2,0), DNO number(1,0), varchar2(15)), where ENO is primary key and DNO is foreign key referencing DEPTNO of table DEPT.
- 2 List out DEPT_NO and department wise total BASIC_PAY for all employees of EMP table.
- 3 To give permission to modify the structure of table STUD to user XYZ.
- 4 Select ENO, ENAME from EMP where the ENAME start with the character 'A'.

[B] Define cursor and write the steps to manage explicit cursor. [06]

OR

[B] Assume that table named TELE has fields TNO Number(2), CNAME varchar2 (20), NCALLS number(4). [06]

Write a PL/SQL code block to read TNO from user and to display the message "Given TNO exists in table TELE" if TNO is found in table TELE otherwise display the message that "Given TNO does not exist in table TELE".